EXPERIMENT 1.4

Mapped Course Outcomes-CO3

CO3: Analyze the different routing techniques.

Aim:- Implementation of Static Routing using n routers.

Objective: How static routing is created , maintained and updated by a network administrator using static route.

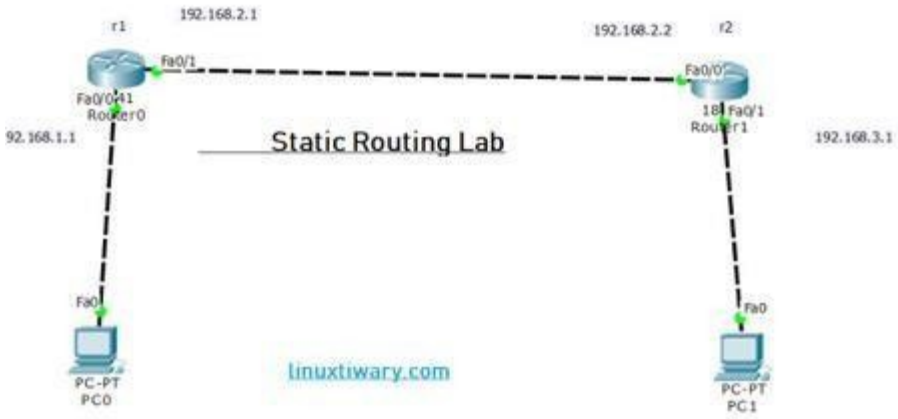
Prerequisites:-

1. An access to CISCO PACKET TRACER .

A static routing table is created, maintained, and updated by a network administrator, manually. A static route to every network must be configured on every router for full connectivity. This provides a granular level of control over routing but quickly becomes impractical on large networks. Routers will not share static routes with each other, thus reducing CPU/RAM overhead and saving bandwidth. However, static routing is not fault-tolerant, as any change to the routing infrastructure (such as a link going down, or a new network added) requires manual intervention. Routers operating in a purely static environment cannot seamlessly choose a better route if a link becomes unavailable. Static routes have an Administrative Distance (AD) of 1, and thus are always preferred over dynamic routes, unless the default AD is changed. A static route with an adjusted AD is called a floating static route.[1]

Step by Step Procedure:

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Configu ration .

Router0 Configuration.

Router#conf terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastEthernet 0/0

Router(config-if)#ip address 10.0.0.1 255.0.0.0

Router(config-if)#no shut Router(config-

if)#exit Router(config)#interface

fastEthernet 1/0

Router(config-if)#ip address 192.168.1.1 255.255.255.0

Router(config-if)#no shut

Router(config-if)#exit

Router(config)#ip route 20.0.0.0 255.0.0.0 192.168.1.1

Router(config)#exit

Router1 Configuration.

Router#conf terminal

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#interface fastEthernet 0/0

Router(config-if)#ip address 20.0.0.1 255.0.0.0

Router(config-if)#no shut Router(config-

if)#exit Router(config)#interface

fastEthernet 1/0

Router(config-if)#ip address 192.168.1.2 255.255.255.0

Router(config-if)#no shut

Router(config-if)#exit

Router(config)#ip route 10.0.0.0 255.0.0.0 192.168.1.2

Router(config)#exit

Further Reading:-

1. https://www.techopedia.com/definition/26161/static-routing

Research Articles

1. Dowdell, J. & Benamar, Nabil. (2015). Static Routing for DTN. Gardner, W.R. 1987. Water content: an overview. International Conference on Measurement of Soil and Plant Water Status. Centennial of Utah State Univ., pp. 7-9.

2. Saini, Himanshi and Pondwal, Veenu, An Analysis of Static and Dynamic Routing Techniques Based on Various Network Performance Parameters (May 2017). The IUP Journal of Telecommunications, Vol. IX, No. 2, May 2017, pp. 20-31, Available at SSRN: https://ssrn.com/abstract=3212546